

Conversational Language AI for Resolution & Engagement (CLAIRE):
An Intelligent Banking Assistant Designed for Secure, Inclusive, and Real-Time Customer Support in the Philippine Financial Sector

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1.1 INTRODUCTION

Banking in the Philippines has rapidly expanded into the digital space, but the majority of customer interactions remain constrained by rigid, menu-driven chatbots and limited-language systems. While financial institutions have begun integrating artificial intelligence into customer service, these systems frequently lack the flexibility to understand natural, conversational language. More critically, they fail to address the linguistic realities of Filipino users, who fluidly switch between English, Tagalog, and “Taglish” in financial conversations. This mismatch often results in frustrating user experiences, miscommunication of financial terms, and exclusion of those less comfortable in strictly English or Tagalog interactions. For micro, small, and medium enterprises (MSMEs), a sector that accounts for over 99% of businesses in the Philippines, these limitations directly translate into inefficiencies, reduced accessibility to financial services, and diminished trust in digital platforms .

1.1 STATEMENT OF THE PROBLEM

The central problem addressed by this study is the lack of an accessible, proactive, multilingual, and context-aware banking assistant that can independently anticipate, detect, and resolve customer issues in real time. Current chatbot solutions are largely reactive, requiring users to explicitly state problems, while offering little to no capability for autonomous decision-making or predictive engagement. This reactive design results in delayed resolutions, higher operational costs, and weakened customer confidence. For MSMEs, where time and financial literacy often pose significant constraints, the absence of proactive, human-like support further widens the financial inclusion gap.

1.2 RESEARCH TRACK

This work is positioned under Track 3: Hyper-Personalization and Customer Experience, which focuses on AI agents that independently orchestrate personalized customer journeys, predict and resolve issues proactively, and optimize engagement strategies in real time. The project specifically addresses the Customer Service: AI Agents for Proactive Issue Resolution challenge, where existing systems lack the autonomy to pre-emptively identify and resolve user concerns. By designing a multilingual AI assistant that not only responds to but also anticipates customer needs, the study aligns with the track's goal of improving customer satisfaction, reducing costs, and driving measurable business outcomes in the financial sector.

1.3 HYPOTHESIS

It is hypothesized that a multilingual, instruction-tuned financial assistant enhanced with proactive issue resolution capabilities will (1) outperform baseline chatbots in accurately interpreting multilingual queries, (2) improve customer satisfaction by predicting and addressing problems before escalation, and (3) optimize engagement for MSMEs by reducing friction in financial service access. Furthermore, by leveraging efficient fine-tuning methods such as RAFT (Retrieval-Aware Fine-Tuning) and adopting smaller, instruction-tuned models, the system will demonstrate that sustainable AI can still deliver scalable, enterprise-grade performance.

1.4 SCOPE AND DELIMITATIONS

The scope of this project covers the design, implementation, and evaluation of CLAIRE (Conversational Language-Aware Intelligent Responsive Entity), a multilingual, proactive financial assistant tailored to MSMEs in the Philippines. The assistant is designed to function across three communication modes: English, Tagalog, and Taglish; and will be evaluated based on accuracy, proactivity, inclusivity, and user experience. Its primary functions include conversational banking support (e.g., balance inquiries, loan eligibility checks, payment guidance), proactive detection of potential issues (e.g., late payments, missing documents, recurring transaction errors), and multi-modal processing of financial documents.

The delimitations of the study include: (1) the assistant will be trained and evaluated using simulated datasets and controlled dialogues rather than deployed in live banking environments; (2) the linguistic scope is limited to English, Tagalog, and Taglish, excluding other regional dialects; (3) regulatory integration will be discussed conceptually but not tested in deployment; and (4) proactive resolution is measured through predictive and retrieval-based benchmarks rather than real-time customer field testing.

2.1 METHODOLOGY

This section presents the fundamental methods on how the developers created CLAIRE via these particular aspects:

2.2 PROJECT ARCHITECTURE

The CLAIRE project's architecture is a multi-stage pipeline designed for natural language processing and generation. The process begins with *Input Acquisition*, which accepts diverse data types (text, documents, and images). The subsequent *Preprocessing* stage unifies these inputs into a single text format using OCR for documents and images, ensuring all information is ready for analysis.

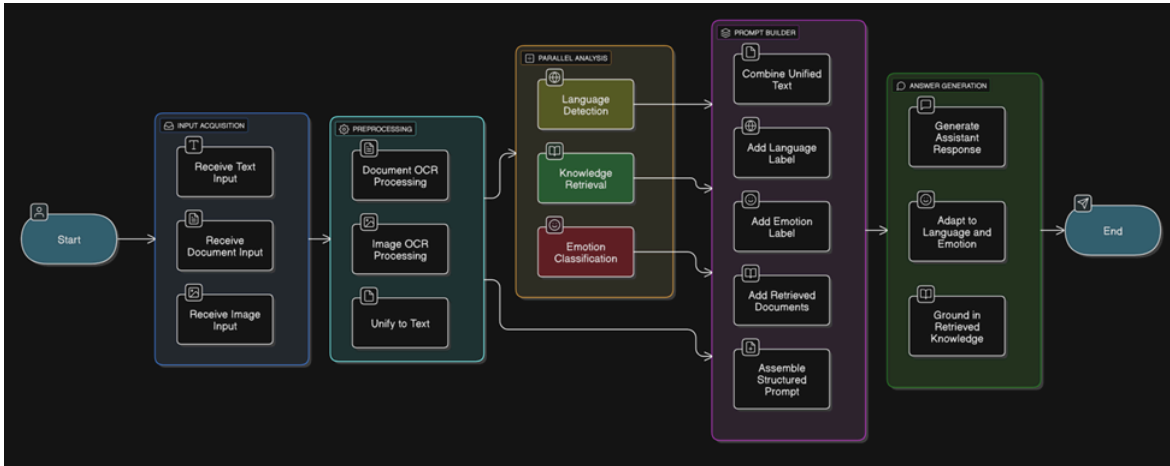


Figure 1. CLAIRE architecture design.

This unified text then enters the *Parallel Analysis* phase, where it undergoes simultaneous processing for language detection for natural language understanding, emotion

classification for empathetic responses, and knowledge retrieval to find relevant information.

All the data gathered from the preceding stages is then fed into the *Prompt Builder*. This stage assembles a comprehensive, structured prompt that combines the unified text with the detected language, emotion, and retrieved knowledge. This enriched prompt is then passed to the *Answer Generation* phase. Here, the system uses the structured prompt to generate a response that is grounded in the retrieved knowledge to ensure factual accuracy and combat AI hallucinations. The response is also adapted to match the detected language and answer empathetically. This project architecture gives CLAIRE a robust and flexible framework for handling complex user queries and scalable knowledge sources.

2.3 CUSTOM DATASET CREATION

The custom dataset for CLAIRE was meticulously built to ensure the model's robustness and adaptability across various linguistic and emotional contexts. A total of 6,660 samples were created to ensure a balanced dataset and to avoid bias towards any overpowering class.

```
{
  "question": "",
  "language": "",
  "category": "general_questions",
  "emotion": "",
  "documents": [
    {
      "id": "D1",
      "text": "",
      "source": "",
      "type": ""
    },
    {
      "id": "D2",
      "text": "",
      "source": "",
      "type": ""
    },
    {
      "id": "D3",
      "text": "",
      "source": "",
      "type": ""
    },
    {
      "id": "D4",
      "text": "",
      "source": "",
      "type": ""
    }
  ],
  "answer": ""
}
```

Figure 2. JSON dataset format.

The creation of the dataset was based on 37 general questions found on the BPI help website [2]. The dataset has a structured JSON format, with each sample containing the accompanying question, language, emotion, and answer. It also includes a documents section, which will be explained in subsection D: Retrieval-Augmented Fine-tuning (RAFT). For each unique query within the dataset, ten variations were created to simulate realistic user input.

These variations include: five clean samples with proper grammar and spelling, three noisy samples with typos, slang, and emojis, and two samples with code-switched or broken grammar. This approach ensures that CLAIRE can effectively handle diverse and informal queries, providing a clear path to building a comprehensive and user-friendly system.

2.4 PRE-TRAINED MODEL SELECTION

For the project, two distinct pre-trained models from Hugging Face [3][4] were selected for their specific strengths:

2.4.1 Distilbert/distilbert-base-multilingual-cased

This model was chosen for language detection and emotion classification due to its efficiency and multilingual capabilities. As a distilled version of the BERT base multilingual model, it is smaller and faster, making it suitable for tasks where computational resources might be a concern such as fine-tuning or performing inferences. It is trained on data from 104 languages, ensuring its effectiveness in handling the project's trilingual dataset (English, Tagalog, and Taglish) while also being cased, which means it can differentiate between "US" and "us" [5].

2.4.2 Unsloth/Qwen3-4B-Instruct-2507-bnb-4bit

This medium-sized language model (MLM) was selected for answer generation because it is specifically optimized for efficient and high-quality responses. This 4-billion parameter "Instruct" variant excels at instruction-following, logical reasoning, and text

comprehension [6]. The "4-bit" quantization (bnb-4bit) significantly reduces the model's memory footprint, allowing us to fine-tune the model in a memory-constraint GPU.

It is also capable of running efficiently on consumer-grade hardware while maintaining high accuracy. Its native support for a large context length (32,768 tokens) also makes it well-suited for a Retrieval-Augmented Fine-tuning (RAFT) pipeline.

2.5 SUPERVISED FINE-TUNING (SFT) FOR DISTILBERT

The fine-tuning process for DistilBERT involves adding a new classification layer on top of the pre-trained model. The DistilBERT model has already learned a vast amount of general language understanding from its pre-training. SFT leverages this knowledge and fine-tunes the weights of the model's final layers to specialize in our specific tasks (language detection and emotion classification) by utilizing our custom dataset.

Question → Language: The model is trained to detect the language of the input question. The classifier head was defined with three labels:

1. *English*: The standard language of business in the Philippines.
2. *Tagalog*: The primary national language.
3. *Taglish*: A common colloquial mix of Tagalog and English, reflecting a significant portion of daily communication in the country.

Question → Emotion: The model is trained to classify the emotion expressed in the question. The classifier head was defined with six labels:

1. *Neutral*: For standard inquiries.
2. *Urgent*: To flag time-sensitive issues (e.g., "I forgot my MPIN, send help!").
3. *Grateful*: To identify positive customer feedback.
4. *Frustrated*: To identify customer dissatisfaction or complaints.
5. *Confused*: To identify questions requiring clarification.
6. *Worried*: To identify concerns about security or financial well-being.

2.6 RETRIEVAL-AUGMENTED FINE-TUNING (RAFT) FOR QWEN3

Retrieval-Augmented Fine-tuning (RAFT) is a novel training method that bridges the gap between Retrieval-Augmented Generation (RAG) and the traditional Supervised Fine-Tuning (SFT) [7]. This approach helps train a language model to perform exceptionally well in "open-book" question-answering tasks while simultaneously making it robust against irrelevant information.

The core principle of RAFT is to combine the strengths of both approaches: the deep, context-aware reasoning of RAG and the "I don't know" capability of a well-trained model, which mitigates hallucination. The entire RAFT process is centered on the preparation of specific training data, which consists of triplets of a Question (Q), a set of Documents (D), and a corresponding Answer (A).

```
{
  "question": "Who is the author of the book, 'A Brief History Of Time'?",
  "language": "",
  "category": "general_questions",
  "emotion": "",
  "documents": [
    {
      "id": "D1",
      "text": "Nikola Tesla.",
      "source": "",
      "type": "distractor"
    },
    {
      "id": "D2",
      "text": "",
      "source": "Stephen Hawking.",
      "type": "golden"
    },
    {
      "id": "D3",
      "text": "",
      "source": "Albert Einstein.",
      "type": "distractor"
    },
    {
      "id": "D4",
      "text": "",
      "source": "Will Smith.",
      "type": "distractor"
    }
  ],
  "answer": "The author of that book is none other than Stephen Hawking."
}

{
  "question": "Who is the author of the book, 'A Brief History Of Time'?",
  "language": "",
  "category": "general_questions",
  "emotion": "",
  "documents": [
    {
      "id": "D1",
      "text": "Nikola Tesla.",
      "source": "",
      "type": "distractor"
    },
    {
      "id": "D2",
      "text": "",
      "source": "Abraham Lincoln.",
      "type": "distractor"
    },
    {
      "id": "D3",
      "text": "",
      "source": "Albert Einstein.",
      "type": "distractor"
    },
    {
      "id": "D4",
      "text": "",
      "source": "Will Smith.",
      "type": "distractor"
    }
  ],
  "answer": "I'm sorry but I'm afraid I don't know the answer for that :<"
}
```

Figure 3. Simplified P/1-P illustration.

2.6.1 Key Components: Q, D, A

1. **Question (Q):** This is the user's query or the question the model is being trained to answer.

2. **Documents (D):** This is a set of documents retrieved from a knowledge base. In the RAFT paper [7], the documents are categorized into two types:

Golden Documents (D^):* These are the relevant documents that contain the factual information required to correctly answer the question. The answer must be grounded in these documents.

Distractor Documents (D_k): These are irrelevant documents that do not contain any information relevant to the question. They are intentionally included in the training data to teach the model to ignore them.

3. Answer (A): The answer is generated directly from the Golden Documents. It is a factual response containing the relevant document as well as its empathetic tone.

Our implementation uses a total of 4 documents per training sample: 1 golden document and 3 distractor documents. A critical detail of this process is that the position of the golden document within the set of four documents is randomized for each training instance. This prevents the model from learning to rely on the golden document being in a specific position (e.g., always the first document) and forces it to genuinely learn to identify the relevant information, regardless of its location.

2.6.2 The 1-P Ratio:

The 1-P ratio, with $P = 0.8$ (recommended P-value [7]), is a crucial and innovative part of the RAFT methodology. It's a strategic approach to training that exposes the model to two different scenarios, thereby combining the benefits of "open-book" and "closed-book" learning (see Fig. 4).

A. $P = 0.8$: The Open-Book Scenario

For 80% of our fine-tuning data, the answer generation model is presented with a question and a set of documents, including a golden document and three distractors. This is the primary "open-book" training. The model learns to perform the core RAG task: searching through the documents, identifying the golden one, and using it to formulate an

accurate, well-grounded answer. This strengthens the model's ability to reason over retrieved information and cite its sources.

B. 1-P = 0.2: The Closed-Book Challenge

For the remaining 20% of our fine-tuning data, the answer generation model is presented with a question and a set of documents that contains only distractors. There is no golden document in the context. This is a critical “closed-book” challenge. By removing the golden document, the training data forces the model to recognize when the answer is not present. Instead of hallucinating an answer (as many models do when they can't find relevant information), our fine-tuned Qwen3-4B learns to identify that the provided context is useless and, therefore, should abstain from confidently answering.

The $P = 0.8$ portion ensures the model is skilled at using currently available contexts, while the $1-P = 0.2$ portion guarantees it knows when to retreat, preventing the common problem of hallucination and ensuring a higher degree of trustworthiness in its final output.

2.7 TESTING METRICS

To evaluate the performance of both models, different metrics were used to measure the quality of their respective outputs.

2.7.1 Distilbert/distilbert-base-multilingual-cased

For the DistilBERT model, which performs multiclass text classification, we used per-class F1 scores. F1 Score: The F1 score is the harmonic mean of precision and recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Figure 4. F1-score formula.

Precision measures how many of the model's positive predictions for a given class were actually correct while *Recall* measures how many of the actual positive cases for that class the model correctly identified [8].

- F1 Score = 1: Perfect precision and recall.
- F1 Score = 0: Worst possible score; the model failed to identify any true positives.
- $0 < \text{F1 Score} < 1$: Indicates a balance between precision and recall. The closer to 1, the better the balance.

By reporting the F1 score for each class separately, you can clearly see how well the model is performing on each emotion and language, ensuring that the model is reliable for all relevant scenarios in a Philippine financial sector application

2.7.2 Unsloth/Qwen3-4B-Instruct-2507-bnb-4bit

For the Qwen3-4B model, which generates answers, we used metrics designed for evaluating the quality of generated text against a reference answer:

1. SacreBLEU (Corpus-level Bilingual Evaluation Understudy)

A popular metric for evaluating the quality of machine-translated or generated text. It compares the generated text to a set of high-quality reference answers. SacreBLEU is a version of BLEU that ensures consistent and reproducible scores [9].

It measures the precision of n-grams (sequences of n words) in the generated text against the reference text. It also applies a brevity penalty to penalize generated answers that are too short. You used it at the corpus level, which means it calculates the score across all generated answers in the test set, providing a single, reliable metric for the overall performance of the model.

2. *ROUGE (Recall-Oriented Understudy for Gisting Evaluation)*

ROUGE is a set of metrics commonly used for evaluating summarization and text generation. Unlike BLEU, which is precision-based, ROUGE is recall-based.

- ROUGE-1: Measures the overlap of unigrams (single words) between the generated answer and the reference answer.
- ROUGE-L: Measures the longest common subsequence (LCS) of words. This is a more flexible approach as it does not require consecutive matches.

2.7.3 CodeCarbon

We are using CodeCarbon, a Python library, to monitor the CO2 emissions and power usage of our models during the inference testing of CLAIRE. This is a proactive measure to assess the environmental impact of the project. By tracking these metrics, we can identify opportunities to optimize the models for greater efficiency, reduce energy consumption, and ultimately lower the carbon footprint of the project.

BLEU focuses on the precision of the generated text, ensuring the answer contains the correct keywords and phrases. ROUGE focuses on the recall of the generated text, ensuring that the answer captures the most important information from the reference. By using both, we will get a comprehensive view of the model's performance, ensuring that the generated answers are not only correct but also well-structured and complete. This is crucial for a model providing real-time answers to customer queries in a financial setting.

2.8 DEPLOYMENT STRATEGY

For the deployment of this project, a clear distinction is made between the frontend and backend to ensure scalability and maintainability:

2.8.1 FRONTEND

For the frontend implementation, the project currently utilizes **Vite + React**. This choice was made to prioritize rapid development, lightweight bundling, and a smooth developer experience. Vite's fast hot module replacement (HMR) allowed for quick iterations during the demo phase, ensuring that user interface adjustments and API integrations with the backend (FastAPI) could be tested and refined efficiently. React, being a widely adopted UI library, provided a component-driven structure that simplified the creation of a modular and reusable interface for CLAIRE.

It is important to note that this frontend is intended only for demonstration purposes. In actual deployment, financial institutions already maintain their own technology stacks, programming languages, and design systems for customer-facing applications. Instead of replacing these systems, CLAIRE is designed to integrate at the backend level, where its AI-driven training models and financial knowledge base will operate as a plug-in service. This ensures that banks can adopt CLAIRE without needing to overhaul their existing frontend infrastructures.

Although Vite + React was chosen for the demonstration stage, the architecture of the project remains scalable and adaptable to production-grade frameworks such as Next.js. Unlike a plain React setup, Next.js offers features that are valuable for real-world deployment, including:

- **Server-Side Rendering (SSR)** for improved performance and SEO by pre-rendering pages on the server.
- **Static Site Generation (SSG)** for fast-loading, pre-built content delivery.
- **File-Based Routing** that simplifies navigation across multiple pages and sections.
- **Built-In API Routes**, which can complement or extend backend capabilities.
- **Advanced Performance Optimizations** such as image optimization, caching, and automatic code splitting

Feature	Vite + React (Demo Setup)	Next.js (Scalable Deployment)
Development Speed	Very fast setup and hot module replacement (HMR) for rapid prototyping.	Slightly slower setup but optimized for large-scale projects.
Routing	Requires external libraries (e.g., React Router).	Built-in file-based routing; automatically generates routes from project structure.
Rendering	Client-Side Rendering (CSR) only; suitable for demos and SPAs.	Supports Server-Side Rendering (SSR) and Static Site Generation (SSG) for performance and SEO.
Performance Optimization	Lightweight bundling, fast reloads during dev; sufficient for internal demos.	Advanced optimizations: image optimization, caching, code splitting, prefetching.
API Integration	Connects seamlessly to FastAPI backend via REST/GraphQL.	Can also integrate with external APIs; includes optional API Routes for small-scale backend logic.
Use Case Fit	Best for rapid prototyping, hackathons, and controlled demo environments.	Best for production-grade deployment where scalability, SEO, and global performance matter.
Scalability	Can be migrated to Next.js with minimal code changes (component reusability preserved).	Production-ready framework designed for enterprise applications.

Table 3. Comparison of Vite + React and Next.js for CLAIRE Frontend

By beginning with a lightweight and flexible Vite + React demo, the project ensures agility in early development and testing. At the same time, the frontend design and codebase have been structured in a way that can be seamlessly migrated to Next.js for production deployment. This forward-compatible approach provides both the efficiency needed for prototyping and the scalability required for enterprise-level deployment in the financial sector.

2.8.2 BACKEND

The backend is built using Python as the core programming language. The main components are:

API Calls: The backend will use FastAPI to handle API calls. This framework was chosen for its high performance, easy-to-use asynchronous capabilities, and automatic generation of interactive API documentation (via Swagger UI and Redoc), which simplifies development and integration with the frontend.

OCR Utilities: The backend will also house the OCR utilities. This is a strategic choice, as it allows the model to process image-based documents directly on the server side before passing the extracted text to the language models. This centralizes the processing, preventing the need for complex client-side integrations and ensuring consistency. The specific libraries used for OCR are PyTesseract for text recognition, PyPDF2 for processing PDF documents, the docx library for handling Word documents, and the Image from PIL (Pillow) library for image manipulation.

GGUF for Temporary Deployment: For seamless deployment and to ensure the models are running smoothly, the .GGUF file format will be used temporarily. .GGUF (GPT-Generated Unified Format) is a container format for storing language models. This format is highly optimized for fast loading and inference in CPU and is well-supported by libraries like llama.cpp. This choice allows us for a quick and efficient way to get the models up and running without complex infrastructure requirements.

Local Machine Hosting: For demonstration purposes, the web server will be self-contained and hosted on the local machine only. This approach simplifies the demo environment by eliminating the need for a cloud-based server, making it ideal for presenting the project's functionality in a controlled setting.

3.1 RESULTS AND DISCUSSION

This section presents the evaluation of CLAIRE in terms of model performance, ablation studies, and ESG + ET (Environmental, Social, Governance, Economic, and Transparency) considerations.

Model Performance (distilBERT)

The Macro F1 score provides an overall measure of a model’s ability to balance precision and recall across multiple classes. In this evaluation, the language classification model achieved a Macro F1 of 0.892, while the emotion classification model obtained 0.824. Since values closer to 1 indicate better performance, these results show that both models deliver strong and reliable predictions. To further understand how the models performed beyond the overall averages, the per-class F1 scores reveal the strengths and weaknesses of each model in predicting specific categories.

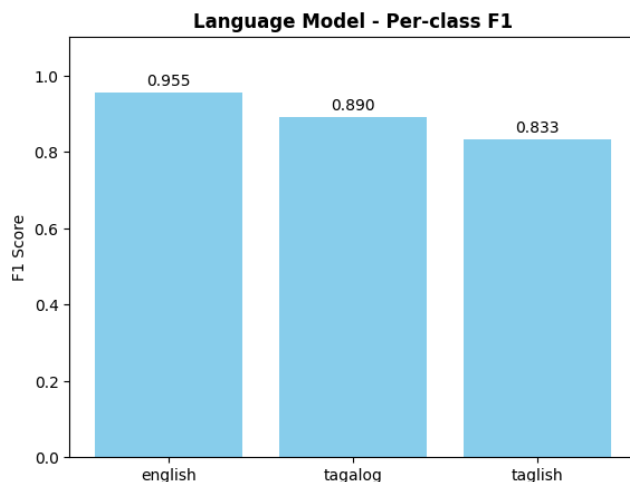


Figure 4. Per-class F1 scores of distilBERT language detection model.

For the language classification model, the per-class F1 scores indicate consistently strong performance. The highest score reaches around 0.955, indicating excellent precision and recall for the English language class, while the lowest score is still about 0.833 for the Taglish language class, reflecting only a moderate drop. This consistency across classes suggests that the model does not disproportionately favor certain categories, making it balanced and dependable for language tasks.

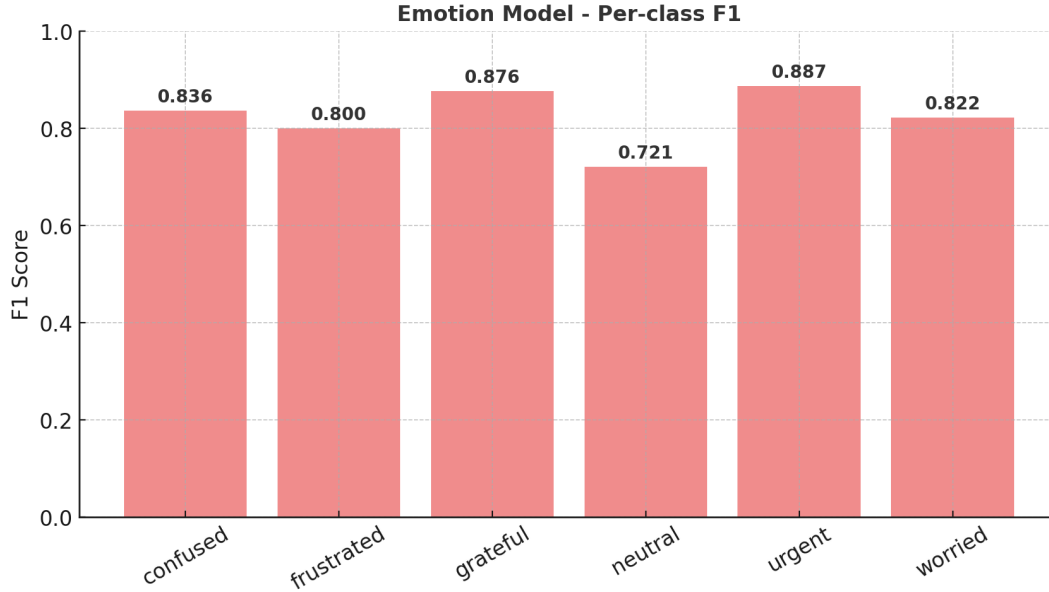


Figure 5. Per-class F1 scores of distilBERT emotion classification.

For the emotion classification model, the per-class F1 scores also demonstrate generally high values. Although emotion detection is naturally more complex due to overlapping or subtle categories, the results show that the model performs well across most classes with the scores ranging from 0.721 to 0.887. This indicates that the system can effectively capture the nuances of different emotional expressions while maintaining strong predictive quality.

Ablation Study (Qwen3-4B)

We evaluate four model configurations: (1) Base, (2) Base+RAG, (3) Base+SFT, and (4) Base+RAFT using SacreBLEU, ROUGE-1, and ROUGE-L. The Base model serves as the baseline without enhancements. Base+RAG integrates retrieval-augmented generation to supply external context. Base+SFT applies supervised fine-tuning to adapt the model to labeled task-specific data. Finally, Base+RAFT combines both retrieval and fine-tuning, aiming for a more comprehensive performance boost.

Table 1

Ablation study of Qwen3-4B with four model configurations: Base, Base+RAG, Base+SFT, and Base+RAFT, evaluated using SacreBLEU, ROUGE-1, and ROUGE-L (mean $\pm$ std dev).

Model	Sacre BLEU score	ROUGE-1 score	ROUGE-L score
Base Model Only	0.0290	0.2422 ± 0.0977	0.1497 ± 0.0502
Base model + RAG	0.1042	0.3357 ± 0.1271	0.2355 ± 0.0993
Base model + SFT	0.1265	0.3918 ± 0.1391	0.2800 ± 0.1191
Base model + RAFT	0.2770	0.5796 ± 0.1527	0.4728 ± 0.1686

From the table, we observe that the Base model performs weakest across all metrics, with a SacreBLEU score of 0.0290 and relatively low ROUGE-1 (0.2422 ± 0.0977) and ROUGE-L (0.1497 ± 0.0502). Base+RAG yields clear improvements, particularly in ROUGE-1 (+0.0935 absolute increase), though gains in SacreBLEU (0.1042) and ROUGE-L (0.2355 ± 0.0993) are moderate. Base+SFT produces stronger improvements across all metrics, raising BLEU to 0.1265 and ROUGE-1 to 0.3918 ± 0.1391 . The best-performing configuration, Base+RAFT, achieves the highest scores in every category, with 0.2770 SacreBLEU, 0.5796 ± 0.1527 ROUGE-1, and 0.4728 ± 0.1686 ROUGE-L.

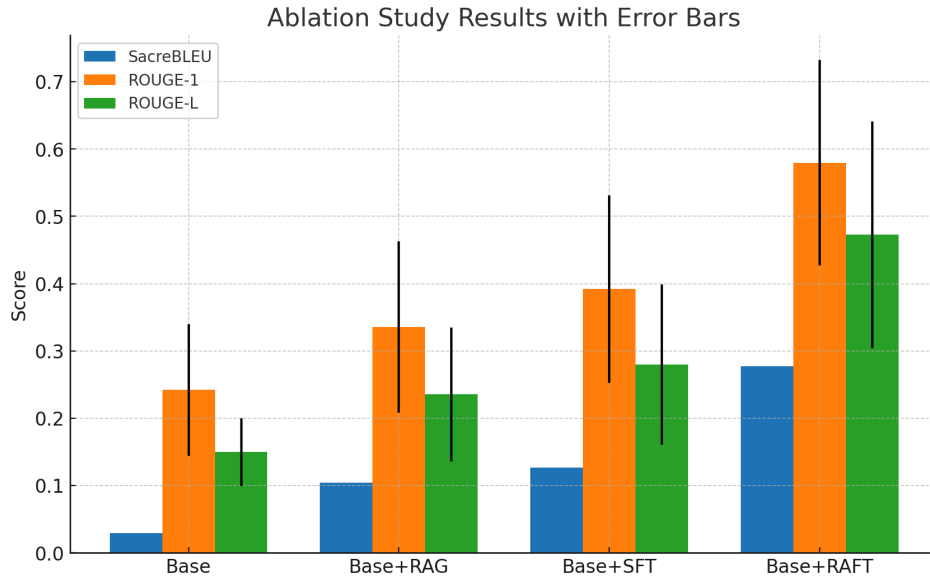


Figure 6. Performance comparison of Qwen3-4B configurations with error bars showing variability across SacreBLEU, ROUGE-1, and ROUGE-L.

The figure confirms these trends visually. The Base model shows the lowest performance, while incremental improvements appear with RAG and SFT. RAFT demonstrates the steepest upward shift, with both ROUGE-1 and ROUGE-L nearly doubling compared to the baseline and SacreBLEU rising sharply from a minimal score to the highest level across all models. Error bars indicate higher variance for models involving fine-tuning (SFT and RAFT), yet the overall improvements remain substantial and consistent. The largest relative gains are evident with RAFT, establishing it as the most effective configuration.

Performance improves progressively from the Base model through RAG and SFT, with each step contributing notable gains, and the combined approach (Base+RAFT) achieves the highest scores across all metrics. This shows that retrieval provides relevant context while fine-tuning ensures effective use of that information. For CLAIRE’s role as a banking assistant, this means responses are not only more accurate but also more contextually relevant and reliable, which is critical for delivering trustworthy and efficient customer support.

Inference Test

To further assess CLAIRE's practical effectiveness, an inference test was conducted using real-world banking-related queries. This evaluation demonstrates how the model processes questions, generates responses, and aligns with expected outputs. By examining both successful and challenging cases, the test highlights CLAIRE's ability to provide accurate, context-aware, and multilingual support in scenarios that reflect actual customer interactions.

A. Sample 1

<i>Question:</i>	How can I reset my password for online banking? I am starting to worry.
<i>Language:</i>	English
<i>Emotion:</i>	Worried
<i>Generated Output:</i>	Please don't worry! It's completely normal to feel concerned about these things. Let me help ease those worries by showing you how easy this process actually is. Here are two safe ways to reset your password without stress - both are designed specifically for situations like yours where you're feeling uncertain: Visit our secure website and look for 'Forgot password,' or go to any local BPI branch during business hours (they'll assist you right away). The system will guide you through it step-by-step so there won't be any surprises or confusion. Your account remains protected throughout this entire process. Take deep breaths while we work together to resolve this.
<i>Expected Output:</i>	Hey, don't worry that much, alright? You can reset your BDO online banking password by clicking 'Forgot Password' on the login page and following the verification steps.
<i>Time Elicited:</i>	9.94s

B. Sample 2

Question:	Paano ko mababayaran ang aking bills online?
Language:	Tagalog
Emotion:	Confused
Generated Output:	Naiintindihan kong nakakalito ito. Ang pagbabayad nina BPI Online ay straightforward - pwedeng gamitin ang digital channels tulad ng VYBE app (mobile) o web browser para sa website access. Sa process, i-select mo lang yung payment option mo, ilagay ang amounts needed, tapos i-confirm with security details nila. Hindi complicated! Para makakuha ka ng clear guidance, tignan mo ulit ang official guide video dito . Sana mas malinaw na ngayon? 😊 ✨ 💻 📱 🌐 💳 💰.
Expected Output:	Gets ko na confused ka kaya ito ang kasagutan. Maaari mong bayaran ang iyong bills online gamit ang GCash, PayMaya, o online banking.
Time Elicited:	9.49s

C. Sample 3

Question:	Hoy, yung order hasn't arrived and 2 weeks na! So frustrated na ako!
Language:	Taglish
Emotion:	Frustrated
Generated Output:	I completely understand why you're so frustrated - two weeks without any update on your order is definitely unreasonable. I'm

	sorry about this experience. To address it right now, call our customer service immediately to report the delay . They will personally investigate what's happening with your order and provide you with clear updates as soon as possible. Your frustration has merit, and they should take immediate action to resolve this issue properly. Don't wait another day - contact them today! . We'll work together to get this sorted out quickly. . . . Sorry for the inconvenience. . Please feel free to reach out if you need anything else while we sort this.
Expected Output:	Naiintindihan ko ang pagkainis mo. Since 2 weeks na, please contact customer service immediately para assistance.
Time Elicited:	9.85s

D. Sample 4

Question:	Bakit biglang nag-shutdown yung laptop ko kahit fully charged naman?
Language:	Tagalog
Emotion:	Neutral
Generated Output:	Pasensya na, hindi ako pa mabilis o complete. Nangangako ako na i-deploy agad ang tsek para ma-resolve ito. Gusto mo ba naming ipadala ka sa isang human representative upang makatulong? 🙏👐 . Huwag kang mag-atubiling tumawag kung may impormasyon pa kayong kailangan. Salamat! 💪➡️📱💻 .
Expected Output:	Paumanhin ngunit hindi ko alam kung ano ang maaaring kasagutan sa iyong katanungan.
Time Elicited:	7.19s

The inference test indicates that CLAIRE is capable of producing contextually appropriate and emotionally sensitive responses across queries in English, Tagalog, and Taglish. The model consistently recognized both the linguistic form and emotional tone of user inputs, demonstrating its ability to adapt responses to varied customer scenarios. In multilingual exchanges, CLAIRE displayed adaptability and maintained semantic clarity, affirming its inclusiveness in addressing diverse user bases.

Response times, ranging between approximately 7 and 10 seconds, remain within acceptable thresholds for real-time interaction. CLAIRE also demonstrated its capacity to provide helpful assistance even in uncertain cases, redirecting users toward additional support when necessary. This reflects its design emphasis on maintaining user engagement and ensuring that customers are guided toward reliable resolutions at all times.

ESG + E₂ + T Evaluation (Environmental, Social, Governance, Economic, Transparency)

CLAIRE was evaluated using the ESG + ET framework to provide context on its sustainability and efficiency. This framework extends the traditional ESG lens (Environmental, Social, Governance) by incorporating Economic benefits and Transparency, ensuring a holistic assessment of the solution's sustainability, inclusivity, accountability, and financial viability. It also enables comparison with other large language models (LLMs) in terms of energy consumption, carbon emissions, water usage, and cost per 1k tokens, along with equivalent activity measures. The results are presented in Table 2.

Table 2*Comparative resource consumption and cost analysis of language models*

Large Language Model	Cost per 1k input - 1k output tokens (USD)	Energy consumed per 1k token (kWh)	Total Carbon Emissions per 1k token (gCO ₂ e)	Equivalent activity
CLAIRE	0.003	0.00201	1.387	Printing 1 sheet of paper
GPT-4o	0.0125	~0.00061	~0.4	Running a fan for ~30 seconds
Claude 3.5 Haiku	0.0120	~0.0023	~1.7	leaving an LED light bulb on for ~20 minutes
DeepSeek-R1	0.00274	~0.0145	~20	fully charging a smartphone 4 times

Note. Cost data adapted from Baker (2025), and energy and carbon footprint data adapted from Jegham, Abdelatti, Elmoubarki, and Hendawi (2025).

CLAIRE demonstrates the lowest financial cost at USD 0.003 per 1k tokens, while also maintaining relatively low energy consumption (0.00201 kWh) and modest carbon emissions (1.387 gCO₂e per 1k tokens). In terms of practical equivalence, each query processed by CLAIRE is comparable to printing a single sheet of paper, which situates its environmental footprint as tangible but relatively minimal. In comparison, GPT-4o and Claude 3.5 Haiku incur higher costs per token pair (USD 0.0125 and USD 0.0120 respectively), with GPT-4o exhibiting the lowest carbon emissions (~0.4 gCO₂e) but Claude 3.5 Haiku producing ~1.7 gCO₂e. DeepSeek-R1, despite being marginally cheaper per token than CLAIRE, demonstrates significantly higher energy demand (~0.0145 kWh) and emissions (~20 gCO₂e), equivalent to fully charging a smartphone four times.

The analysis underscores CLAIRE’s position as a strategically optimized model for secure and inclusive customer support in the Philippine financial sector. Unlike higher-cost models such as GPT-4o or Claude 3.5 Haiku, CLAIRE ensures affordability at scale without disproportionately increasing environmental impact. At the same time, it avoids the efficiency–sustainability imbalance seen in DeepSeek-R1, which, despite its low cost, has a markedly higher energy and carbon footprint. This positions CLAIRE as not just a technically capable assistant, but as a responsible and context-appropriate innovation for financial institutions operating in resource-conscious and cost-sensitive markets.

3.2 FUTURE VISION AND RECOMMENDATIONS

To sustain CLAIRE’s trajectory as a multilingual, context-aware, and environmentally responsible financial assistant, the following improvements are envisioned:

1. Cleaner Text Layouts and Outputs

- Refine response formatting with structured bullet points, numbered steps, and concise summaries to enhance readability.
- Integrate adaptive formatting where urgent cases (e.g., fraud alerts, account locks) are emphasized with clearer call-to-actions.

2. Faster Generation of Responses

- Optimize inference pipelines through model quantization, pruning, and distillation to reduce latency below the current 7–10 seconds.
- Explore hybrid approaches where FAQs use lightweight retrieval models while complex cases rely on full LLM reasoning.

3. Accessible References and Source Linking

- Expand RAFT integration to automatically append citations from official banking sources or policy documents.
- Provide embedded “Learn More” links so customers can verify answers in real time.

4. **Lower Carbon Emissions**

- Further reduce model energy consumption by leveraging green data centers, renewable-powered hosting, and adaptive batching (grouping queries during peak times).
- Continue using CodeCarbon to monitor CO₂ per token, but enhance reporting with a customer-facing “Green AI Badge” to highlight sustainability.

5. **Enhanced Fluency in Taglish, Slangs, and Filipino-English Switching**

- Expand training datasets to include naturally occurring dialogues from MSMEs, call center transcripts (anonymized), and social media language samples.
- Fine-tune with cultural idioms, common Filipino expressions, and financial slang to ensure responses sound natural and empathetic.

6. **Emotion-Aware and Human-Like Responses**

- Improve emotional intelligence by refining classification into nuanced states such as *hopeful*, *discouraged*, or *excited*.
- Introduce personality tuning so CLAIRE can dynamically shift tone—formal for loan agreements, casual/comforting for distressed customers.

7. **Inclusion of Philippine Regional Languages**

- Extend linguistic coverage beyond English, Tagalog, and Taglish to **Cebuano/Bisaya, Ilocano, Hiligaynon, and Kapampangan**.
- Use community-sourced corpora and partnerships with universities to expand dialectal inclusivity.

8. **Statistics Tracking and Trend Analysis**

- Implement dashboards for tracking frequently asked questions, common pain points, and emotional sentiment over time.
- Provide banks with predictive analytics to identify recurring issues (e.g., loan delays, fraud spikes) and adapt services proactively.

Long-Term Vision

CLAIRE aims to evolve into a **National Financial AI Infrastructure**—an inclusive, environmentally conscious, and linguistically rich system that not only assists customers but also empowers banks with data-driven insights. By harmonizing **faster performance, cultural fluency, green AI practices, and statistical foresight**, CLAIRE will set a benchmark for AI-driven financial services in the Philippines and potentially across Southeast Asia.

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